Hidden sources of phosphorus in the typical American diet: does it matter in nephrology?

Elevated serum phosphorus is a major, preventable etiologic factor associated with the increased cardiovascular morbidity and mortality of dialysis patients. An important determinant of serum phosphorus is the dietary intake of this mineral; this makes dietary restriction of phosphorus a cornerstone for the prevention and treatment of hyperphosphatemia. The average daily dietary intake of phosphorus is about 1550 mg for males and 1000 mg for females. In general, foods high in protein are also high in phosphorus. These figures, however, are changing as phosphates are currently being added to a large number of processed foods including meats, cheeses, dressings, beverages, and bakery products. As a result, and depending on the food choices, such additives may increase the phosphorus intake by as a much as 1 g/day. Moreover, nutrient composition tables usually do not include the phosphorus from these additives, resulting in an underestimate of the dietary intake of phosphorus in our patients. Our goal is to convey an understanding of the phosphorus content of the current American diet to better equip nephrologists in their attempt to control hyperphosphatemia.